

ECO 209 Test 2

2025-01-15

Last name:

First name:

Student ID:

Instructions:

1. Read each question carefully before attempting to answer.
2. Write legibly and preferably in pen.
3. Explain all of your answers. Show all necessary steps. Be concise.
4. Make all extra assumptions that you consider necessary but make sure to state them clearly.
5. Make sure to label all your diagrams appropriately.
6. You have to hand in the exam sheet.
7. DO NOT turn this page until instructed.

Time: 100 minutes

Total Score: $30+15+10+35 = 90$ points; Pages: 20
ONLY Non-Programmable Calculators is ALLOWED

1. **(30 pts)** Consider the consumption model discussed in Ch.16, the consumer has the following utility function:

$$U = \ln(c) + \beta \ln(c')$$

where c is the consumption in the current period, c' is the consumption in the future period and β is the discount factor. The consumer inherits f amount of wealth in the current period and the consumer can take f' amount of asset to the future period. Also, the consumer earns an exogenous income of y and y' in the current and future period respectively.

In this economy, the government collects revenue using lump-sum tax (i.e., a tax unrelated to income or consumption) and consumption tax (i.e., a tax depends on the level of consumption). The government collects T and T' in the current and future period respectively. The consumption tax will deduct τc from current period income and $\tau' c'$ from future period income. Here, τ and τ' represent the rates of the consumption tax in current period and in future period.

- (a) **(3 pts)** Set up the budget constraints. Present the current period budget constraint, the future period budget constraint, and the lifetime budget constraint.
- (b) **(4 pts)** Derive the Euler equation using either the substitution method or the Lagrangian method.
- (c) **(4 pts)** Determine the optimal values of c and c' as functions of exogenous variables and parameters. Denote these optimal values as c_c^* and $c_c'^*$.
- (d) **(6 pts)** For the parts below, assume $\tau = \tau' = 0.13$ and the consumer has income of \$11,000 today and \$16,000 in the future. The lump-sum tax are $T = T' = 3000$. Also, the consumer has an initial asset (f) of \$2,000. The consumer can save or borrow at $r = 0.05$. The discount factor (β) is equal to 0.95.

What is the value of assets in the future period and the amount of consumption tax collected in each period?

- (e) **(3 pts)** Now suppose the government decides to replace the consumption tax with only lump-sum tax (i.e., $\tau = \tau' = 0$). Based on previous records, the government increases the lump-sum tax by an amount equal to the consumption tax collected in part (c) (i.e., $\Delta T = \tau c_c^*$ and $\Delta T' = \tau c_c'^*$). The new lump-sum tax can be defined as $T_e = T + \Delta T$ and $T'_e = T' + \Delta T'$.

Modify your budget constraints from part (a) by incorporating ΔT and $\Delta T'$ into the existing budget constraint.

- (f) **(6 pts)** Under this new policy, determine the optimal values of c and c' using the provided numbers and your previous answers. How does the optimal value of f' compare to its value before the policy change?
- (g) **(4 pts)** The government assumed that replacing the consumption tax with a lump-sum tax would not affect consumption, based on the concept of Ricardian equivalence. Does Ricardian equivalence apply in this case? Why or why not? Using your answer above, explain in 3–5 sentences.

Solution

a) (It is ok if students plug in number from part d))

Current period budget constraint: $(1 + \tau)c = y + f - f' - T$

Future period budget constraint: $(1 + \tau')c' = y' + (1 + r)f' - T'$

Combine the two, we have the lifetime budget constraint:

$$(1 + \tau)c + \frac{(1 + \tau')c'}{1 + r} = y - T + \frac{y' - T'}{1 + r} + f$$

b) Note that $c = c(c') = \frac{1}{1 + \tau} \left(\bar{X} - \frac{(1 + \tau')c'}{1 + r} \right)$ where $\bar{X} = y - T + \frac{y' - T'}{1 + r} + f$

Then, $\frac{\partial c(c')}{\partial c'} = -\frac{(1 + \tau')}{(1 + \tau)(1 + r)}$

Taking derivative of the utility function, we have

$$\frac{1}{c} \frac{\partial c(c')}{\partial c'} + \beta \frac{1}{c'} = 0$$

Sub in the $\frac{\partial c(c')}{\partial c'}$ and rearrange the equation, we have

$$\frac{c'}{c} = \frac{1 + \tau}{1 + \tau'}(1 + r)\beta$$

*Lagrangian method would probably cleaner and easier for students who have already learn the method.

c) Rearrange the Euler equation:

$$c' = \frac{1 + \tau}{1 + \tau'}(1 + r)\beta c$$

Plug it into the budget constraint

$$\begin{aligned}
 (1 + \tau)c + \frac{(1 + \tau')}{1 + r} \frac{1 + \tau}{1 + \tau'} (1 + r)\beta c &= y - T + \frac{y' - T}{1 + r} + f = \bar{X} \\
 \implies [(1 + \tau) + (1 + \tau)\beta]c &= (1 + \tau)(1 + \beta)c = \bar{X} \\
 \implies c^* &= \frac{\bar{X}}{(1 + \tau)(1 + \beta)}
 \end{aligned}$$

Sub it back to the arranged Euler equation:

$$c'^* = \frac{1}{1 + \tau'} (1 + r) \frac{\beta}{(1 + \beta)} \bar{X}$$

d) Using the future period budget constraint and the optimal c' ,

$$(1 + \tau') \frac{1 + r}{1 + \tau'} \frac{\beta}{(1 + \beta)} \bar{X} = (1 + r) \frac{\beta}{(1 + \beta)} \bar{X} = y' + (1 + r)f' - T'$$

Plug in the number and solve for f' .

$$\begin{aligned}
 (1 + 0.05) \frac{0.95}{(1 + 0.95)} \left(11000 - 3000 + \frac{16000 - 3000}{1 + 0.05} + 2000 \right) \\
 = 16000 + (1 + 0.05)f' - 3000 \\
 f' = -1,477.41
 \end{aligned}$$

The consumption tax collects are:

$$\tau c^* = 0.13 (10157.00) = 1320.41$$

$$\tau c'^* = 0.13 (10131.61) = 1317.11$$

e) The new budget constraint would become:

$$(1 + 0)c + \frac{(1 + 0)c'}{1 + r} = y - (T + \Delta T) + \frac{y' - (T' + \Delta T')}{1 + r} + f$$

- f) From the early calculation, we have the $\Delta T = \tau c^* = 1320.41$ and $\Delta T' = \tau c'^* = 1317.11$ The $\bar{X}$ would be updated to be

$$\bar{X} = 11000 - (3000 + 1320.41) + \frac{16000 - (3000 + 1317.11)}{1.05} + 2000 = 19806.15$$

We can re-use the equation we have earlier:

$$c^* = \frac{\bar{X}}{(1 + \tau)(1 + \beta)} = \frac{1}{1.95} 19806.15 = 10157.00$$

$$c'^* = \frac{1 + r}{1 + \tau'} \frac{\beta}{(1 + \beta)} \bar{X} = \frac{1.05 \times 0.95}{(1.95)} 19806.15 = 10131.61$$

The f' would become

$$(1.13)10131.61 = 16000 + 1.05f' - 3000$$

$$f' = -1477.41$$

- g) No, the Richardian equivalence would not apply in this case, although the end result is the same in this particular case. The Richardian equivalence suggests that government borrowing to finance expenditures (change of the timing of tax) does not affect the consumption, because consumer would adjust their profile using the financial market as well. In our case, the government does not change the timing of the tax but the type of tax. So, although the consumption does not change at the end (the optimal consumption in part f and part d are the same), it is not because of the Richardian equivalence.

*If the answer is based on the optimal c only and have correctly stating the fact about the Richardian equivalence, we may give close to full mark (or even full mark).

2. (15 pts) In the Bathtub model, the labour force $\bar{L}$ can exist in two states: employed (E_t) or unemployed (U_t) during period t . The evolution of unemployment over time is described by the equation:

$$\Delta U_{t+1} = \bar{s}E_t - \bar{f}U_t$$

where ΔU_{t+1} is the change in employment, $\bar{s}$ is the job separation rate and $\bar{f}$ is the job-finding rate.

The economist has presented labour market data in the table below.

	Separation Rate (%)	Finding Rate (%)	labour Force (millions)
2010	0.25	4.5	153.9
2013	0.35	3.8	155.4
2015	0.30	4.1	157.1
2018	0.20	3.3	162.1

- (a) (4 pts) Derive the steady state unemployment rate.
- (b) (2 pts) What is the natural rate of unemployment in each of these years ?
- *You do not need to show the formula for each individual number; however, partial marks may be awarded if the final answer is incorrect but reasonably close.
- (c) (3 pts) What is the number of *employment* in each of these years? Show the general formula.
- *You do not need to show the formula for each individual number; however, partial marks may be awarded if the final answer is incorrect but reasonably close.
- (d) (3 pts) How would the natural rate of unemployment be affected if the government implements programs such as job training initiatives or employment placement services that increase the job finding rate? Explain using the natural rate of unemployment formula, and clearly state all the assumptions necessary to support your conclusion.

- (e) **(3 pts)** What would happen to the unemployment rate in the short run (ΔU_{t+1}) if the economy experiences a recession characterized by an increase in the separation rate ($\bar{s}$) and a decrease in the job finding rate ($\bar{f}$)?

Solution

- (a) **(4 pts)** In steady state, $\Delta U_{t+1} = 0 = \bar{s}E - \bar{f}U$. Rearrange and plug in $E = \bar{L} - U$

$$\bar{s}(\bar{L} - U) = \bar{f}U$$

Solve for U :

$$U = \frac{\bar{s}\bar{L}}{\bar{s} + \bar{f}} \implies u = \frac{U}{\bar{L}} = \frac{\bar{s}}{\bar{s} + \bar{f}}$$

- (b) **(2 pts)**

	Separation Rate (%)	Finding Rate (%)	labour Force (millions)	Nature rate of unemployment (%)
2010	0.25	4.5	153.9	5.26
2013	0.35	3.8	155.4	8.34
2015	0.30	4.1	157.1	6.82
2018	0.20	3.3	162.1	5.71

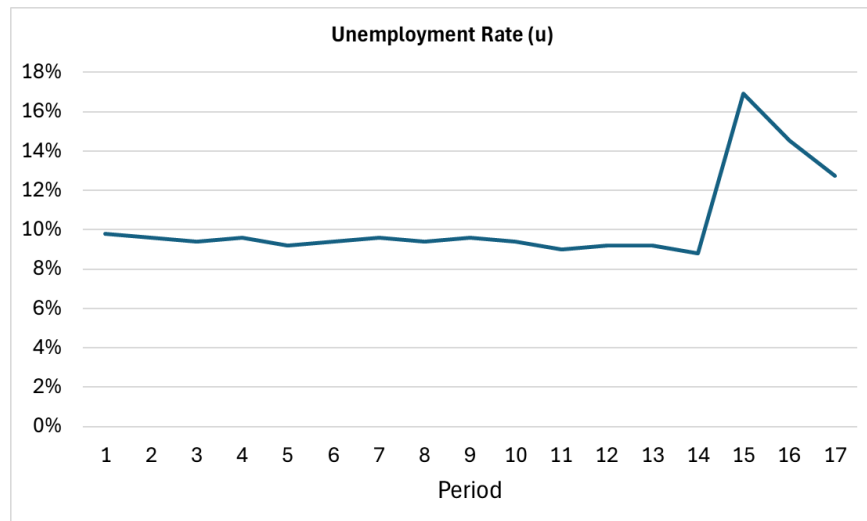
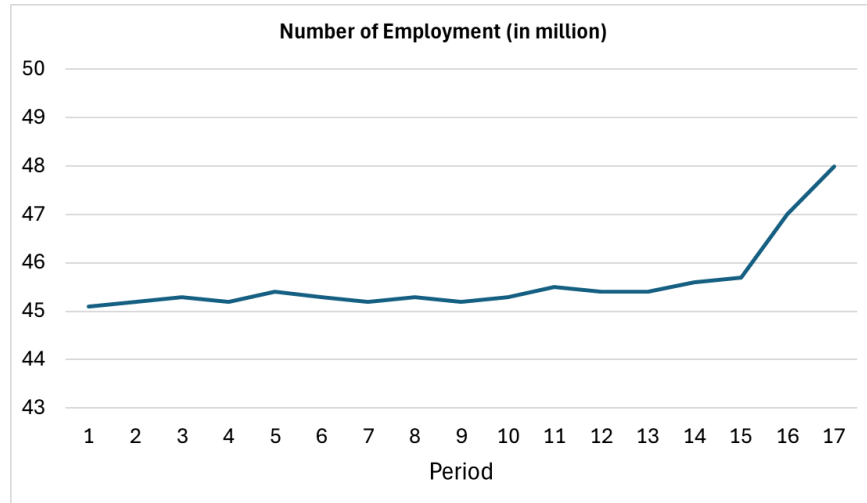
- (c) **(3 pts)**

The employment rate is $1 - u$. Multiply the labour force by it, we can get the number of employment.

	Separation Rate (%)	Finding Rate (%)	labour Force (millions)	Nature rate of unemployment (%)	Number of Employment
2010	0.25	4.5	153.9	5.26	145.80
2013	0.35	3.8	155.4	8.34	142.29
2015	0.30	4.1	157.1	6.82	146.39
2018	0.20	3.3	162.1	5.71	152.84

- (d) From the steady state equation, an increase in $\bar{f}$ would lead to a decline in the nature rate of unemployment.
- (e) If $\bar{s}$ rises and $\bar{f}$ falls, the $\Delta U_{t+1} = \bar{s}E_t - \bar{f}U_t$ would increase as a result.

3. (10 pts) Shown below are figures showing the number of employed individuals and the unemployment rate for a hypothetical economy.



Let E represent the number of employed individuals, U the number of unemployed individuals, L the number of labour force, and u the unemployment rate.

- (a) **(2 pts)** How can the unemployment rate u and labour force L be defined in terms of the symbols given above?
- (b) **(3 pts)** Why does the unemployment rate experience a sharp increase in period 15? Explain using the definition provided above.
- (c) **(5 pts)** Given the conclusion you have in part b), trace a rough graph of the number of labour force and the number of unemployment over time.

Solution

(a)

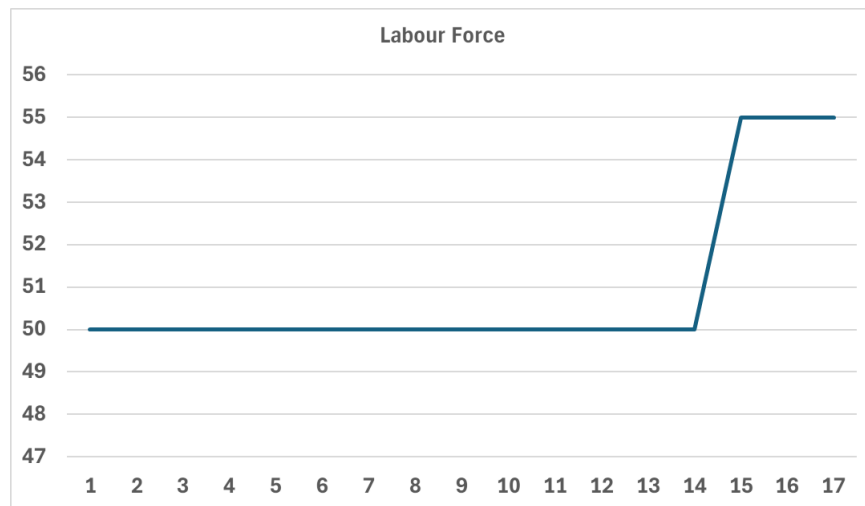
$$u = \frac{U}{L}$$
$$L = U + E$$

(b) By using the definition above,

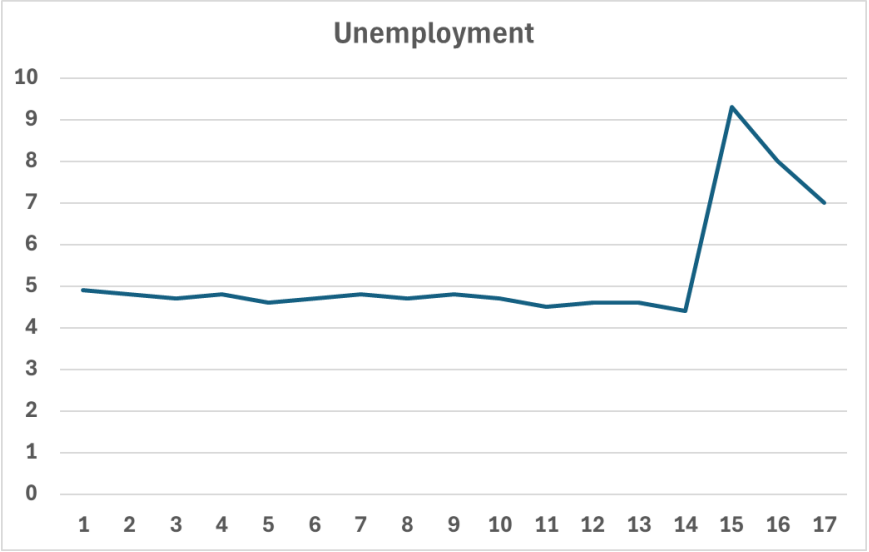
$$u = \frac{U}{L}$$

The only way for this to increase is through an increase in U . However, we observe that total employment (E) remains unchanged in period 15 and increases afterward. This implies that L must also increase as a result, given that $L = U + E$.

(c) *The labour force would have a shape rise in period 15. After that, there could be flat (like in the graph below), and slightly moving up or down as well.



*If the labour force figure shown a upward trend after period 15, the unemployment may not fall to match with the unemployment rate figure.



4. **(35 pts)** This question relates to the combined Romer and Solow growth model discussed in Chapter 6. Alton, a closed economy without government spending, features a production function:

$$Y_t = A_t^{3/4} K_t^{1/3} L_{y,t}^{2/3}$$

Here, Y_t represents the output, K_t denotes the capital, $L_{y,t}$ stands for the labour employed and A_t stand for TFP (or stock of ideas) in period t . Alton's capital accumulation follows the standard function:

$$K_{t+1} = (1 - \delta)K_t + I_t$$

In this equation, δ represents the depreciation rate, and I_t represents the investment in period t . The accumulation of ideas following:

$$\Delta A_{t+1} = \bar{z} L_{a,t}$$

where $\bar{z}$ is the productivity of idea creation.

The sum of numbers of workers ($L_{y,t}$) and researchers ($L_{a,t}$) equal to the total population: $L_{y,t} + L_{a,t} = N_t$. The labour supply is inelastic, with the economy consistently providing N_t units of labour at any given wage. The population is growing at the rate of $\bar{n}$. Meanwhile, there are ℓ proportion of the population work in the idea production. Alton's population save a proportion $\bar{s}$ of their output for investment each period.

- (a) **(3 pts)** Convert the production to intensive form.
- (b) **(2 pts)** Convert the formula in part a) into the growth rate.
- (c) **(4 pts)** Show and discuss why the growth rate of output per capita (g_y) and capital per capita (g_k) are equal on the balance growth path?
- (d) **(4 pts)** Show that the growth rate of ideas g_A is determined by constants on the balance growth path.

- (e) **(3 pts)** Express g_y as a function of g_A on the balanced growth path.
- (f) **(3 pts)** Find the balanced growth path of *output per worker* as a function of A_t and other exogenous variables and parameters.
- (g) **(4 pts)** The Alton government is considering implementing measures such as low-cost childcare and tax credits to encourage population growth, as the current total fertility rate has reached an all-time low.

Draw the transition in output per person from the balanced growth path before the policy change to after the policy change. Label all axis and key points of interest. Provide a explanation in 3-4 sentences.

- (h) **(12 pts)** (separate from part g) A portion of Alton's population consists of refugees from a neighboring country. These refugees have fully integrated into Alton's economy, contributing to the workforce in both the research and production sectors in the same proportions as Alton's native population. At period T , the war ends, and half of the refugees decide to return to their home country (ℓ proportion of them are in the idea sector).

Draw the transition in *population*, *output per person* and *total consumption* from the balanced growth path before the policy change to after the policy change. Label all axis and key points of interest. Provide a explanation in 3-4 sentences.

Solution

(a) Divide the production function by N and use $L_{a,t} = (1-\ell)\bar{N} \implies$

$$\frac{Y_t}{N_t} = y_t = A_t^{3/4} \frac{K_t^{1/3} L_{y,t}^{2/3}}{N_t^{1/3} \bar{N}_t^{2/3}}$$

$$y_t = A_t^{3/4} k_t^{1/3} (1-\ell)^{2/3}$$

(b)

$$g_{yt} = \frac{3}{4}g_{At} + \frac{1}{3}g_{kt}$$

(c) Plug in $I_t = sY_t$

$$K_{t+1} - K_t = \Delta K_{t+1} = sY_t - \delta K_t$$

Divide both side by K_t ,

$$\frac{\Delta K_{t+1}}{K_t} = g_K = s \frac{Y_t}{K_t} - \delta = s \frac{Y_t/L_t}{K_t/L_t} - \delta = s \frac{y_t}{k_t} - \delta$$

Now, recall that $k_t = \frac{K_t}{L_t}$ and apply the growth rate approximate formula:

$$g_k = g_K - g_L$$

Next, plug in the g_K above and the fact that $g_L = \bar{n}$

$$g_k = \frac{\Delta k_{t+1}}{k_t} = s \frac{y_t}{k_t} - \delta - \bar{n}$$

On the BGP, $g_{k,t}$ is a constant, therefore, the right-hand side must also be constant. Given that $\bar{s}$, $\bar{n}$ and δ are constants, $\frac{y_t}{k_t}$ must be constant. This condition is met only if y_t and k_t are growing at the same rate. If y_t were growing at a faster rate, the ratio $\frac{y_t}{k_t}$ would increase each period. Therefore, it follows that $g_k^* = g_y^*$.

(d) **(3 pts)** Researchers are used to produce new ideas:

$$g_{At} = \frac{\Delta A_{t+1}}{A_t} = \frac{\bar{z}L_{a,t}}{A_t}$$

Rearrange the the formula,

$$A_t = \frac{1}{g_{A,t}} \bar{z}L_{a,t}$$

On the BGP, $g_{At} = g_{At}^*$, therefore, only the term $L_{a,t}$ is a variable.
Apply the growth rate formula,

$$g_{A,t} = g_{La} = \bar{n}$$

Now, we know that g_{At} is a constant.

(e) **(3 pts)** Plug in $g_{At} = \bar{n}$, $g_K^* = g_Y^*$ into the g_y equation above

$$g_y^* = \frac{3}{4}\bar{n} + \frac{1}{3}g_y^* \implies g_y^* = \left(\frac{3}{2}\right) \left(\frac{3}{4}\right) \bar{n} = \frac{9}{8}\bar{n}$$

For the long-run combined model, this equation pins down the growth rate of output and the growth rate of output per person

*Either using $\bar{n}$ or g_A is fine.

- (f) **(4 pts)** Use the growth rate equation derived earlier. Now, we know that g_y^* is a constant along a balanced growth path. This implies

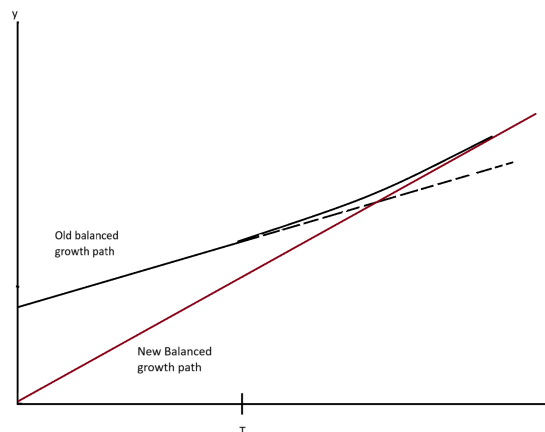
$$\frac{k_t^*}{y_t^*} = \frac{\bar{s}}{g_y^* + \delta + \bar{n}}$$

Rearrange and plug it this : $k_t^* = \frac{\bar{s}}{g_y^* + \delta + \bar{n}} y_t^*$ into the production function:

$$y_t^* = A_t^{3/4} \left(\frac{\bar{s}}{g_y^* + \delta + \bar{n}} y_t^* \right)^{1/3} (1 - \ell)^{2/3}$$

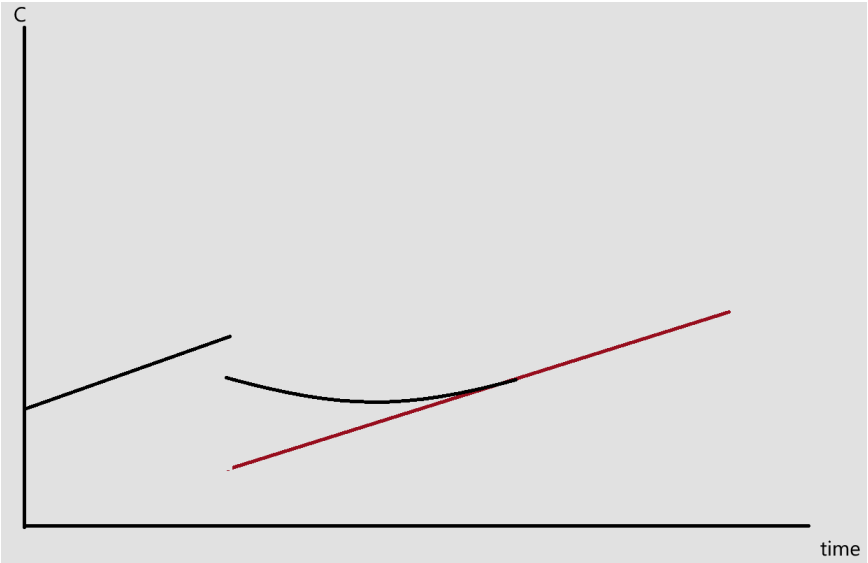
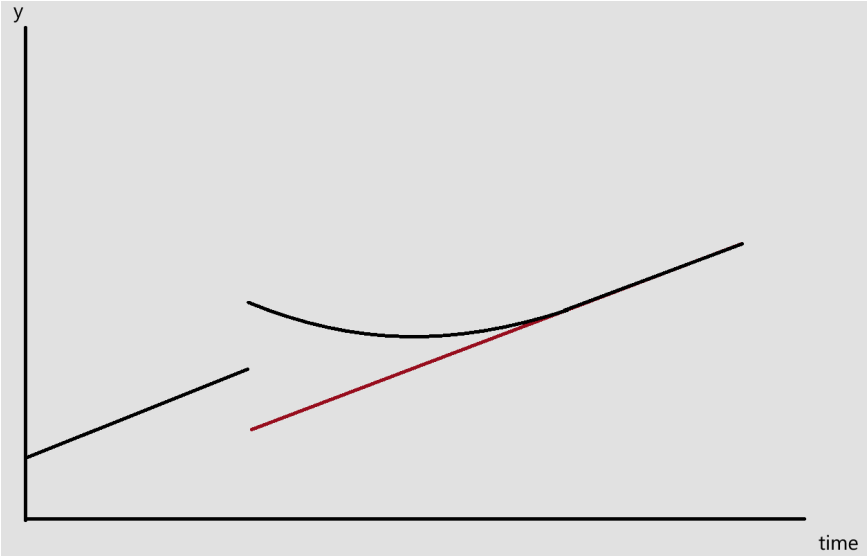
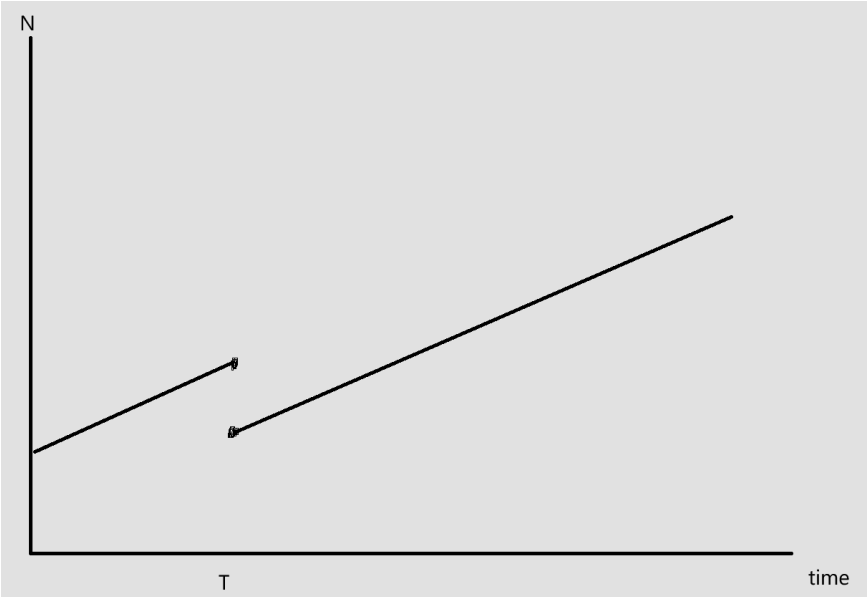
$$\Rightarrow y_t^* = \left[A_t^{3/4} \left(\frac{\bar{s}}{g_y^* + \delta + \bar{n}} \right)^{1/3} \right]^{3/2} (1 - \ell) = A_t^{9/8} \left(\frac{\bar{s}}{g_y^* + \delta + \bar{n}} \right)^{1/2} (1 - \ell)$$

- (g) In this scenario, the government policy would result in an increase in $\bar{n}$. This increase in $\bar{n}$ would lead to a higher growth rate for both g_y , causing the slopes of BGP of y to become steeper than before. However, the new balanced growth path (BGP) would be initially below the original BGP for a certain period because both g_y and $\bar{n}$ have increased. Over time, the new BGP would surpass the old BGP and ultimately rise above it, as A_t grows at a faster rate. In the graph, we assume that the growth from the increase in $\bar{n}$ outpace the slowdown from the other variables (e.g., g_k).



(h) In this case, the N falls right away and it doesn't distort the proportion of worker in the idea and in the production sector. Since the $\bar{n}$ did not change, there is a discrete fall at period T but the slope is unchanged. Due to this, the total output would fall, but the output per capita would go the opposite direction. The output per capita $y_t = A_t^{3/4} k_t^{1/3} (1-\ell)^{2/3}$ would rise because k increase as a result of fall in population. Therefore, we see a discrete jump in y_t . However, there would be a drop in total consumption $C = (1-s)Y$ at Period T since the total output is a function of L_y which fall as a result of people moving out the country. The slope of the BGP is unchanged for all three variables. The BGP for y would shift downward as a result of the decrease in A due to the decrease in N as the economy adjusting its k . Similar effect for the total consumption (recall that $C = (1-s)Y = (1-s)yN$). Initially, the growth rates of y and Y would be lower because both g_A and g_k decrease. The decline in g_k occurs because y_t/k_t decreases as the population falls, given that $\frac{y_t}{k_t} = \frac{A^{3/4}(1-\ell)^{2/3}}{k_t^{2/3}}$. Meanwhile, g_A decreases due to a smaller population size relative to its stock of ideas.

In the graph, this is illustrated by a case where the growth rate turns negative (if y/k is sufficiently low, g_k can become negative), then gradually increases and converges to the balanced growth path (BGP). However, growth does not necessarily have to turn negative; as long as the growth rate slows down initially and then accelerates toward the BGP, the path is valid.



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